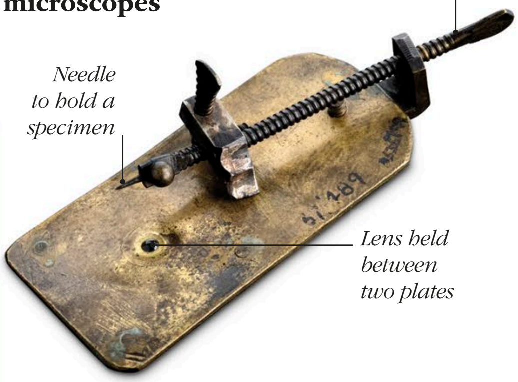
Microscopes

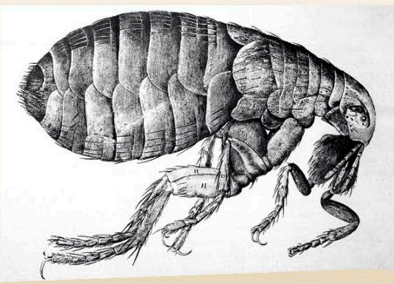
Nobody used to know what causes diseases. In the 1860s, the French chemist Louis Pasteur (see pp.244–245) proved that diseases are caused by tiny organisms, called bacteria, which are too small to be seen with the unaided eye. The journey toward this "germ theory" began around 250 years earlier with the invention of the microscope, which let people see microorganisms for the first time.

Replica of one of Leeuwenhoek's microscopes



EARLY STEPS

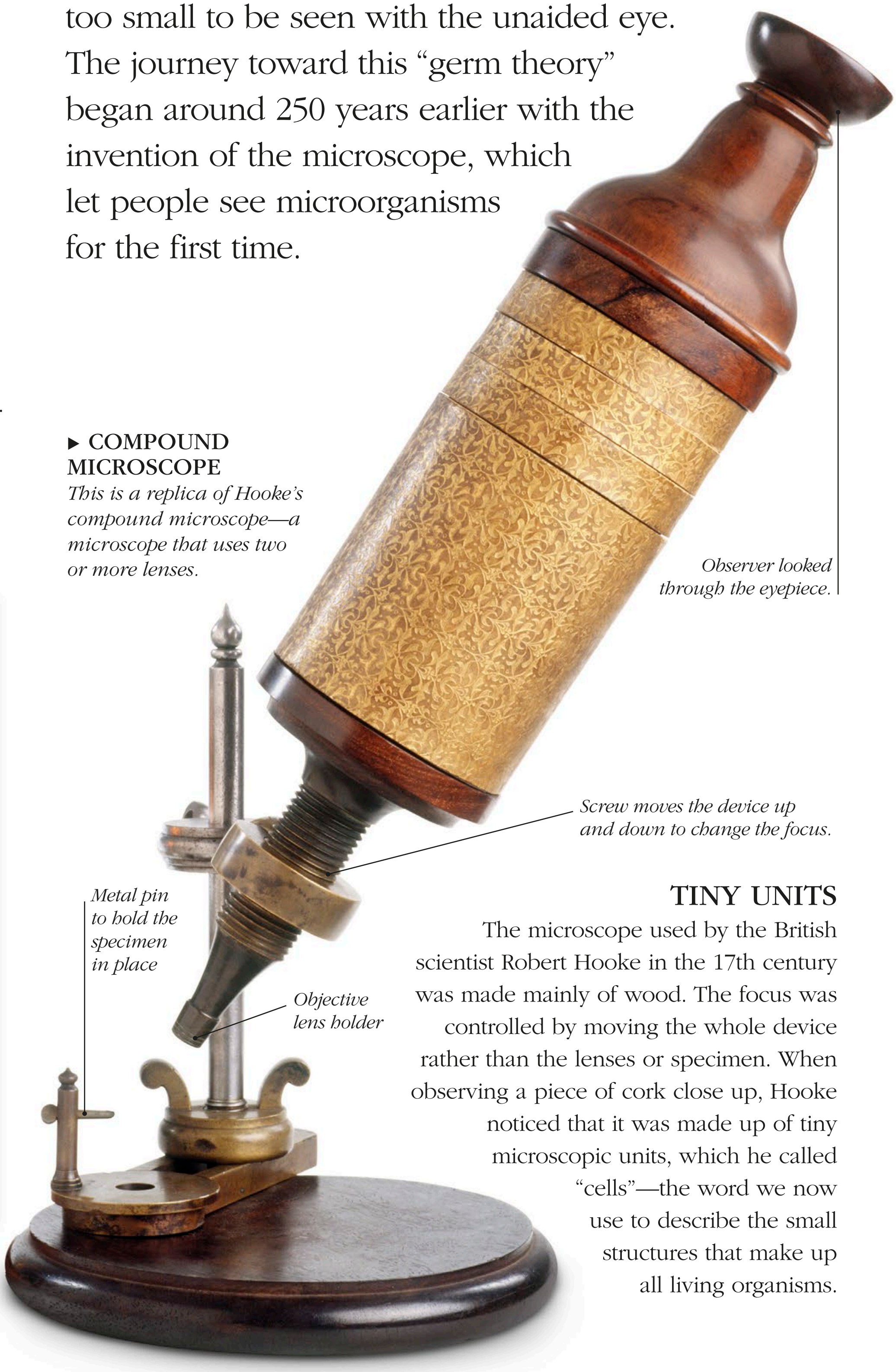
In the 1590s, the Dutch lens makers Hans and Zacharias Jansen combined lenses in a tube to create the first microscopes. In the following century, the Dutch scientist Anton Van Leeuwenhoek built more powerful microscopes, becoming the first person to observe single-celled microorganisms. His device, which had just a single lens, could magnify objects up to 270 times.



MICROSCOPIC LIFE

In the 17th century, scientists began to use microscopes more widely. In 1665, the British scientist Robert Hooke published a book, *Micrographia*, which contained the first illustrations of specimens, such as plants and tiny insects (including the flea above), as seen through a microscope.

► **COMPOUND MICROSCOPE**
This is a replica of Hooke's compound microscope—a microscope that uses two or more lenses.



TINY UNITS

The microscope used by the British scientist Robert Hooke in the 17th century was made mainly of wood. The focus was controlled by moving the whole device rather than the lenses or specimen. When observing a piece of cork close up, Hooke noticed that it was made up of tiny microscopic units, which he called "cells"—the word we now use to describe the small structures that make up all living organisms.